



Assignment 4

Course Details

Course Title: Data Communications

Couse Code: CSE 350

Section: 01

Submitted To

Md. Khalid Mahbub Khan

Lecturer

Department of Computer Science and Engineering

Submitted By

Farhatun Nahar Priya

Student ID: 2023-1-60-202

Date of Submission: 17th November 2025

Introduction

In this assignment, I design, configure, and verify a multi router network using only Class B IP addressing. In the real world, networks are often divided into multiple subnets connected through routers to ensure efficient routing, scalability and security.

Here, I implement a network consisting of three routers, each connected to its own LAN. The routers are interconnected via serial DCE links, forming a triangular topology that provides multiple routing paths between networks. Static routing is used to manually define routes to each destination network, which helps in understanding routing mechanisms and packet flow across an internet network.

Network Topology Overview

The topology consists of

- 3 Routers (Router0, Router1, Router2)
- 3 LAN networks, each containing 2 PCs and 1 switch
- 3 serial DCE connections between routers
- Class B IP addressing only

IP Addressing

Device	Interface	IP Address	Subnet Mask	Network
PC0	Fa0	169.100.0.1	255.255.0.0	169.100.0.0
PC1	Fa0	169.100.0.2	255.255.0.0	169.100.0.0
Router0	Fa0/0	169.100.0.254	255.255.0.0	169.100.0.0
Router0	S2/0	139.100.0.1	255.255.0.0	139.100.0.0
Router0	S3/0	149.100.0.1	255.255.0.0	149.100.0.0
PC2	Fa0	179.100.0.1	255.255.0.0	179.100.0.0
PC3	Fa0	179.100.0.2	255.255.0.0	179.100.0.0
Router1	Fa0/0	179.100.0.254	255.255.0.0	179.100.0.0
Router1	S2/0	149.100.0.2	255.255.0.0	149.100.0.0
Router1	S3/0	159.100.0.1	255.255.0.0	159.100.0.0
PC4	Fa0	189.100.0.1	255.255.0.0	189.100.0.0
PC5	Fa0	189.100.0.2	255.255.0.0	189.100.0.0
Router2	Fa0/0	189.100.0.254	255.255.0.0	189.100.0.0
Router2	S2/0	139.100.0.2	255.255.0.0	139.100.0.0
Router2	S3/0	159.100.0.2	255.255.0.0	159.100.0.0

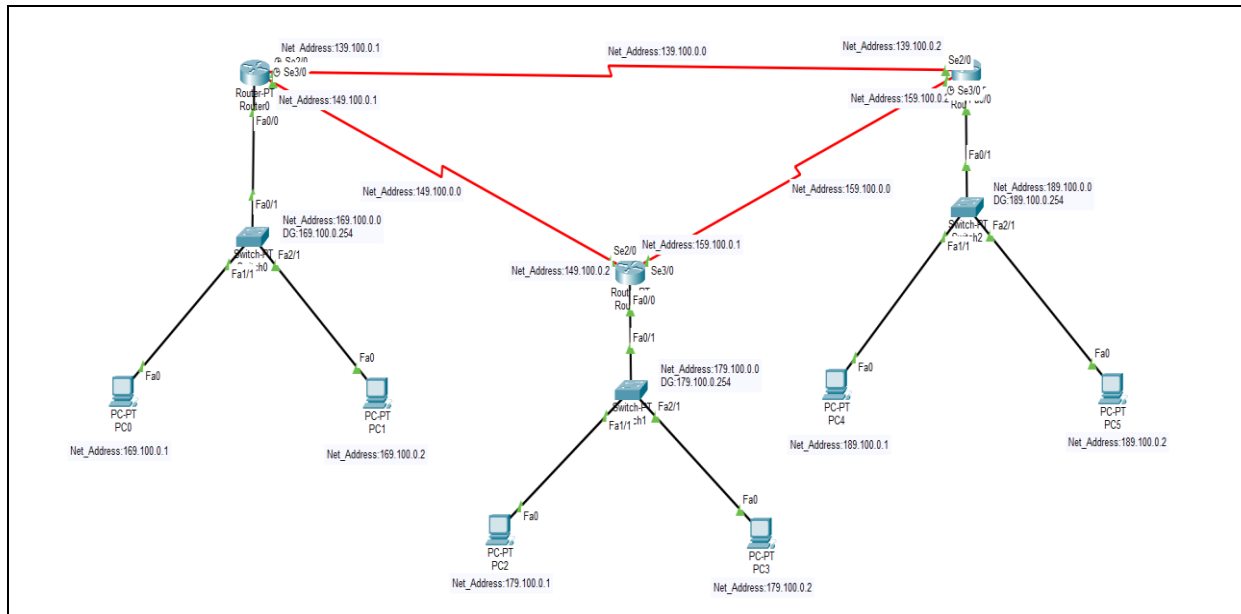


Figure: Network diagram with Static Router Protocol

Router Configurations

For Router 0:

Se2/0

```
Router>en
Router#config t
Router(config)#int se2/0
Router(config-if)#ip address 139.100.0.1 255.255.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int se2/0
Router(config-if)#ip address 139.100.0.1 255.255.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial2/0, changed state to down
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
```

se3/0

```
Router(config)#int se3/0
Router(config-if)#ip address 149.100.0.1 255.255.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

```
Router(config)#int se3/0
Router(config-if)#ip address 149.100.0.1 255.255.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
```

fa0/0

```
Router(config)# int fa0/0
Router(config-if)#ip address 169.100.0.254 255.255.0.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

```
Router(config)# int fa0/0
Router(config-if)#ip address 169.100.0.254 255.255.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
```

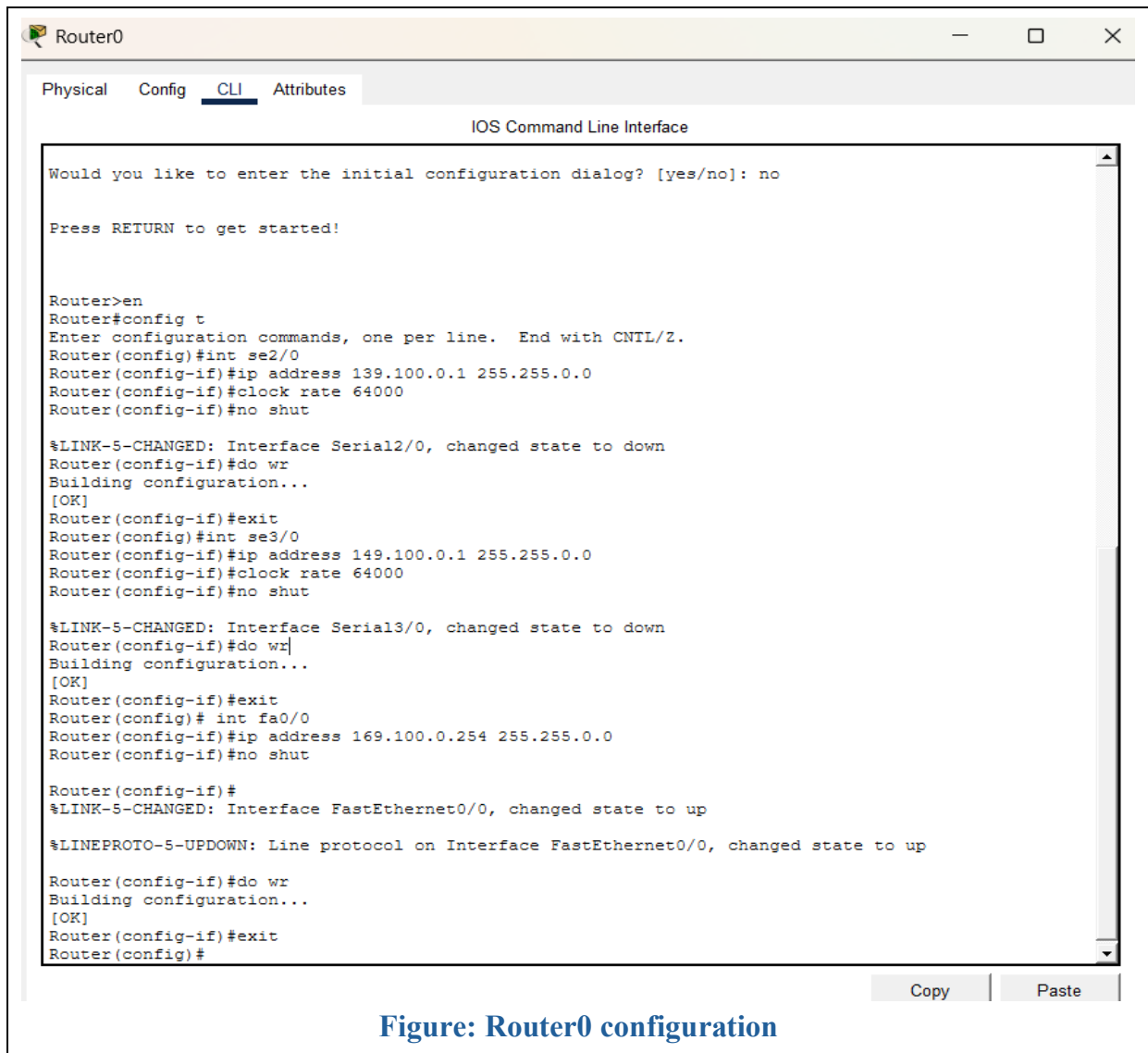


Figure: Router0 configuration

Router0 (Static Configuration)

```
Router>en
```

```
Router#config
```

```
Router(config)#ip route 159.100.0.0 255.255.0.0 149.100.0.2
```

```
Router(config)#ip route 179.100.0.0 255.255.0.0 149.100.0.2
```

```
Router(config)#ip route 189.100.0.0 255.255.0.0 149.100.0.2
```

```
Router(config)#exit
```

```
Router>en
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 159.100.0.0 255.255.0.0 149.100.0.2
Router(config)#ip route 179.100.0.0 255.255.0.0 149.100.0.2
Router(config)#ip route 189.100.0.0 255.255.0.0 149.100.0.2
Router(config)#exit
```

For Router 1

se2/0

```
Router>en
Router#config t
Router(config)#int se2/0
Router(config-if)#ip address 149.100.0.2 255.255.0.0
Router(config-if)#no shut
Router(config-if)#exit
```

```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int se2/0
Router(config-if)#ip address 149.100.0.2 255.255.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

Router(config-if)#exit
```

se3/0

```
Router(config)#int se3/0
Router(config-if)#ip address 159.100.0.1 255.255.0.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

```
Router(config)#int se3/0
Router(config-if)#ip address 159.100.0.1 255.255.0.0
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
```

fa0/0

```
Router(config)#int fa0/0
Router(config-if)#ip address 179.100.0.254 255.255.0.0
```

```
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

```
Router(config)#int fa0/0
Router(config-if)#ip address 179.100.0.254 255.255.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
```

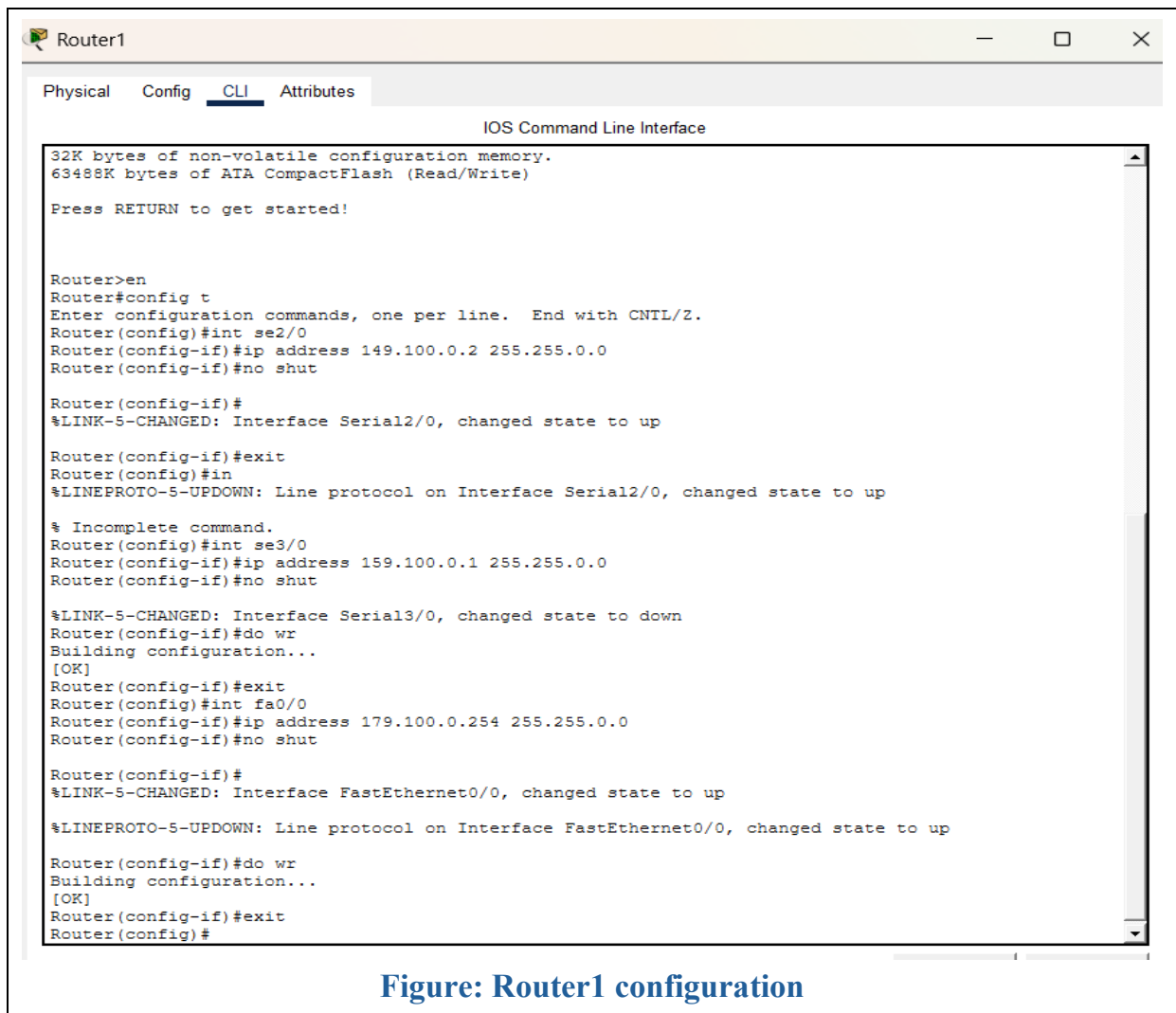


Figure: Router1 configuration

Router1 (Static Configuration)

```
Router(config)#ip route 139.100.0.0 255.255.0.0 159.100.0.2
Router(config)#ip route 169.100.0.0 255.255.0.0 159.100.0.2
```

```
Router(config)#ip route 189.100.0.0 255.255.0.0 159.100.0.2
Router(config)#exit
```

```
Router(config)#ip route 139.100.0.0 255.255.0.0 159.100.0.2
Router(config)#ip route 169.100.0.0 255.255.0.0 159.100.0.2
Router(config)#ip route 189.100.0.0 255.255.0.0 159.100.0.2
Router(config)#exit
```

For Router 2

se2/0

```
Router>en
Router#config t
Router(config)#int se 2/0
Router(config-if)#ip address 139.100.0.2 255.255.0.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int se 2/0
Router(config-if)#ip address 139.100.0.2 255.255.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
Router(config)#
```

Se3/0

```
Router(config)#int se 3/0
Router(config-if)#ip address 159.100.0.2 255.255.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```



```
Router(config)#int se 3/0
Router(config-if)#ip address 159.100.0.2 255.255.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial3/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
```

fa0/0

```
Router(config)#int fa0/0
Router(config-if)#ip address 189.100.0.254 255.255.0.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

```
Router(config)#int fa0/0
Router(config-if)#ip address 189.100.0.254 255.255.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
```

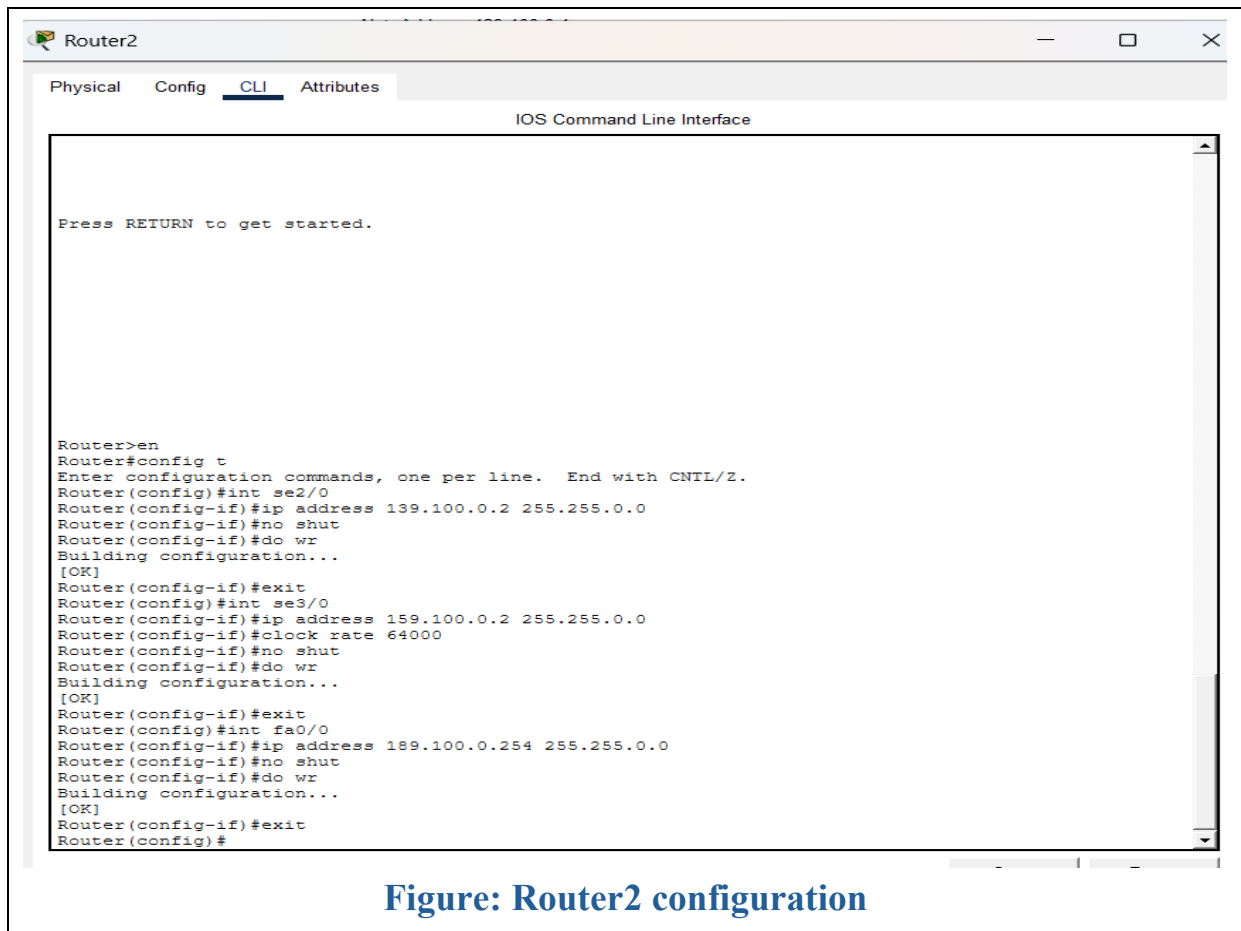


Figure: Router2 configuration

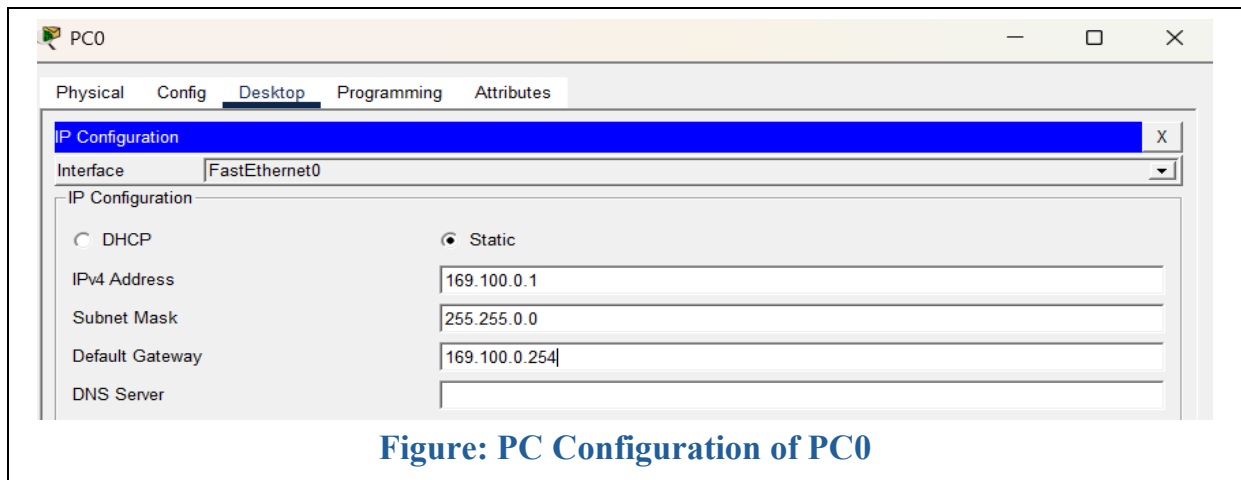
Router2 (Static Configuration)

```
Router(config)#ip route 149.100.0.0 255.255.0.0 139.100.0.1
Router(config)#ip route 169.100.0.0 255.255.0.0 139.100.0.1
Router(config)#ip route 179.100.0.0 255.255.0.0 139.100.0.1
Router(config)#exit
```

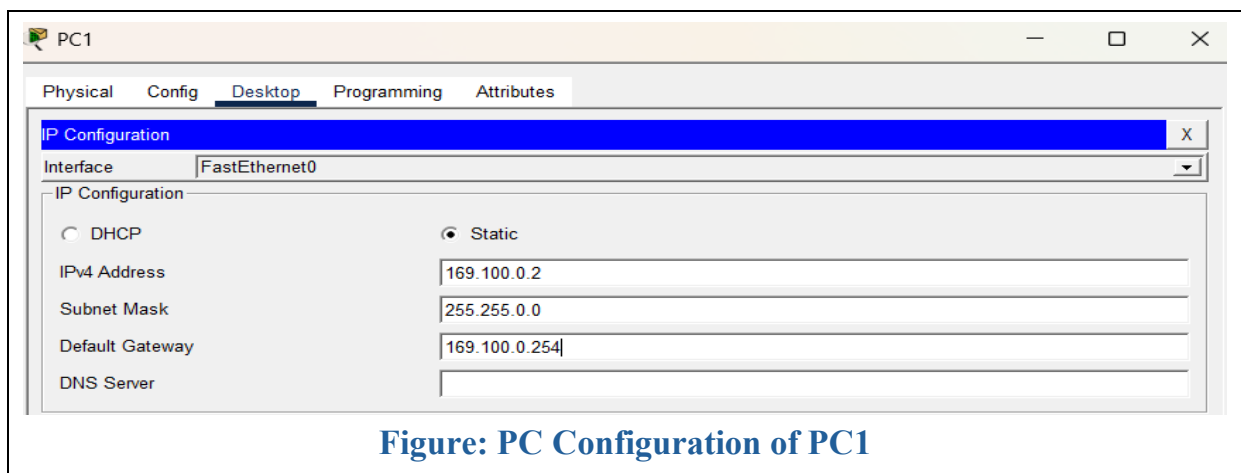
```
Router(config)#ip route 149.100.0.0 255.255.0.0 139.100.0.1
Router(config)#ip route 169.100.0.0 255.255.0.0 139.100.0.1
Router(config)#ip route 179.100.0.0 255.255.0.0 139.100.0.1
Router(config)#exit
```

Device Configuration

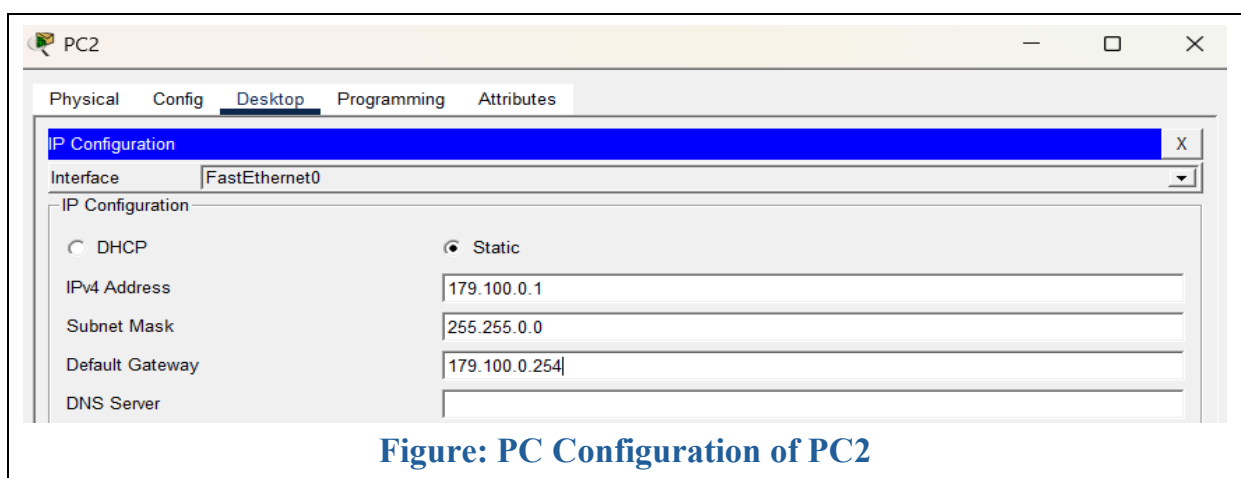
For PC 0



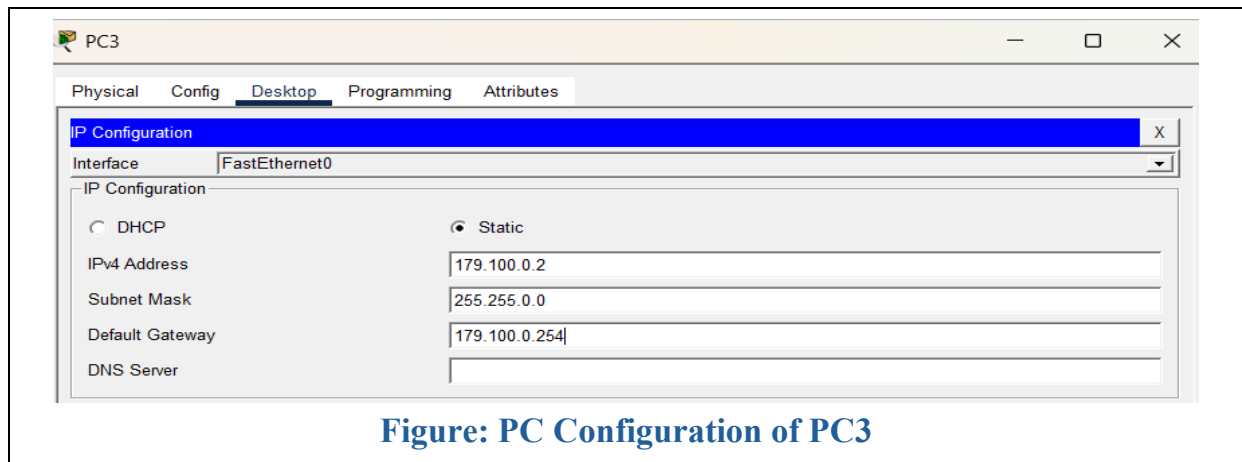
For PC 1



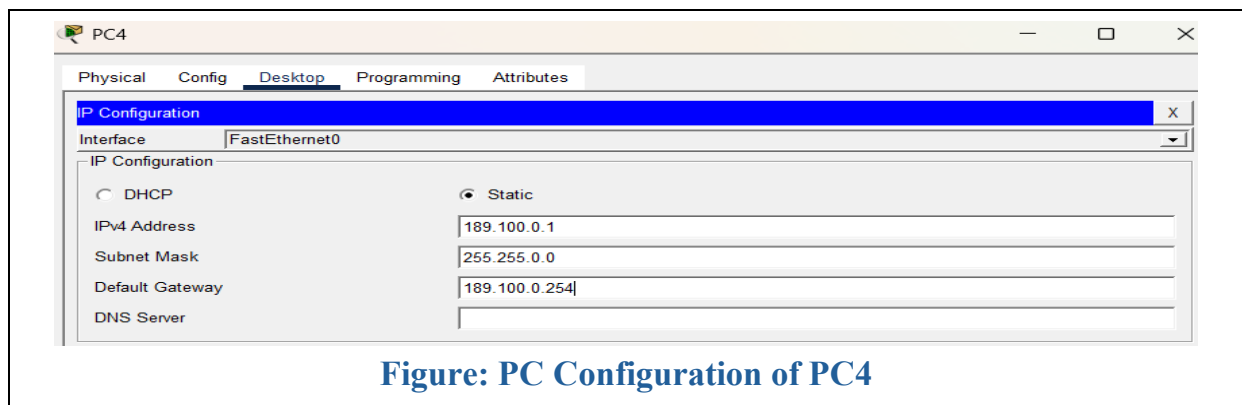
For PC 2



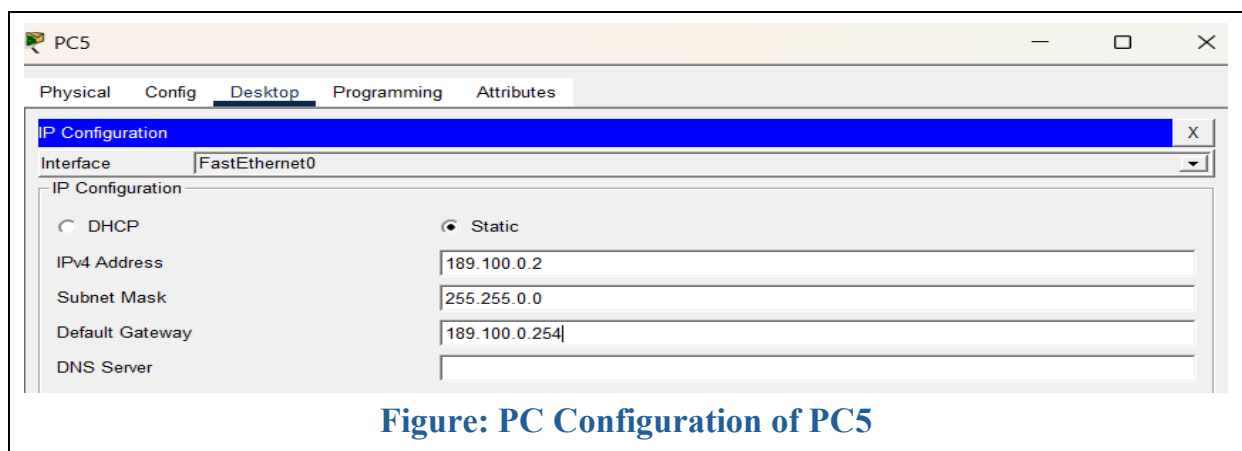
For PC 3



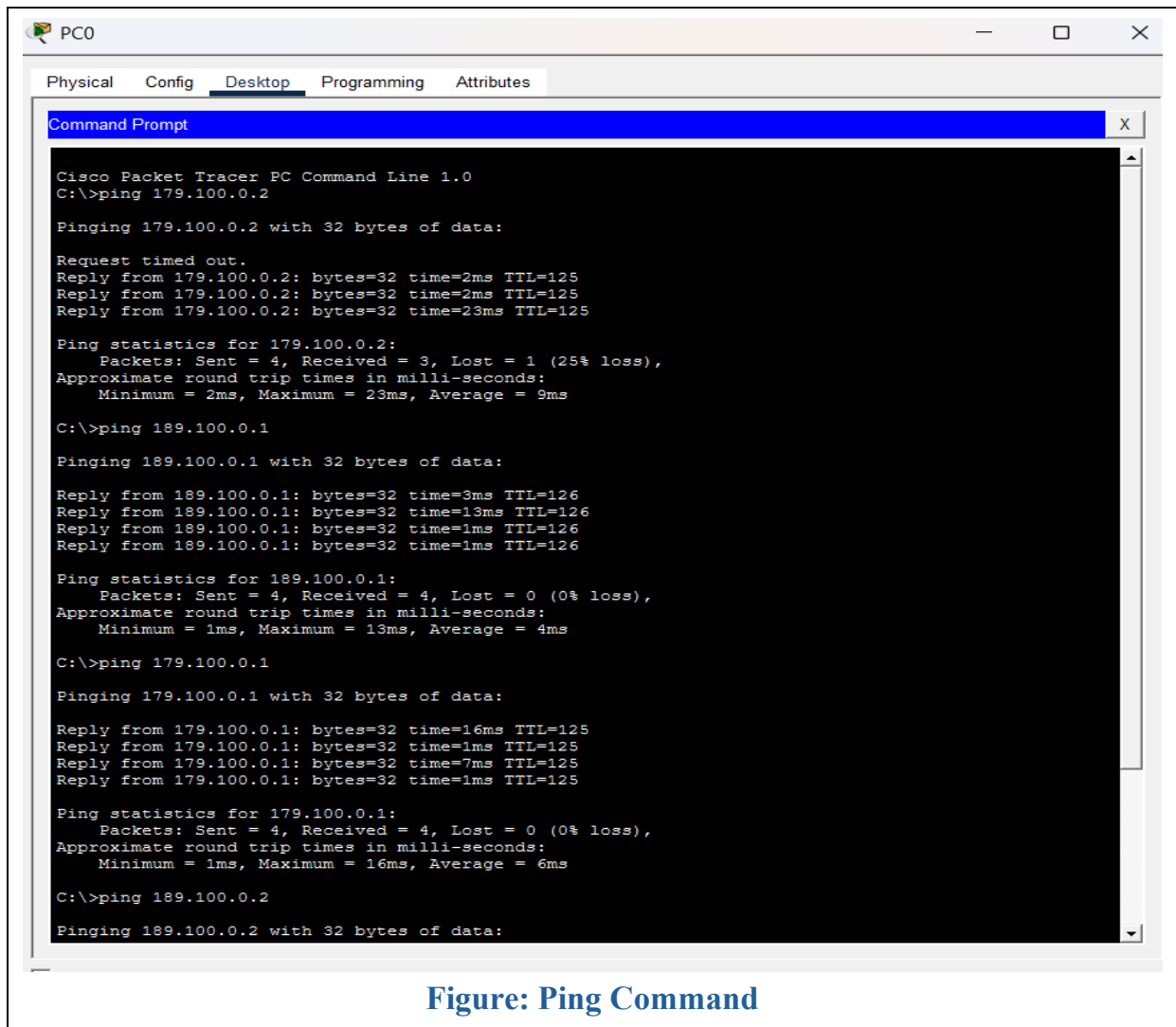
For PC 4



For PC5



Ping Communication



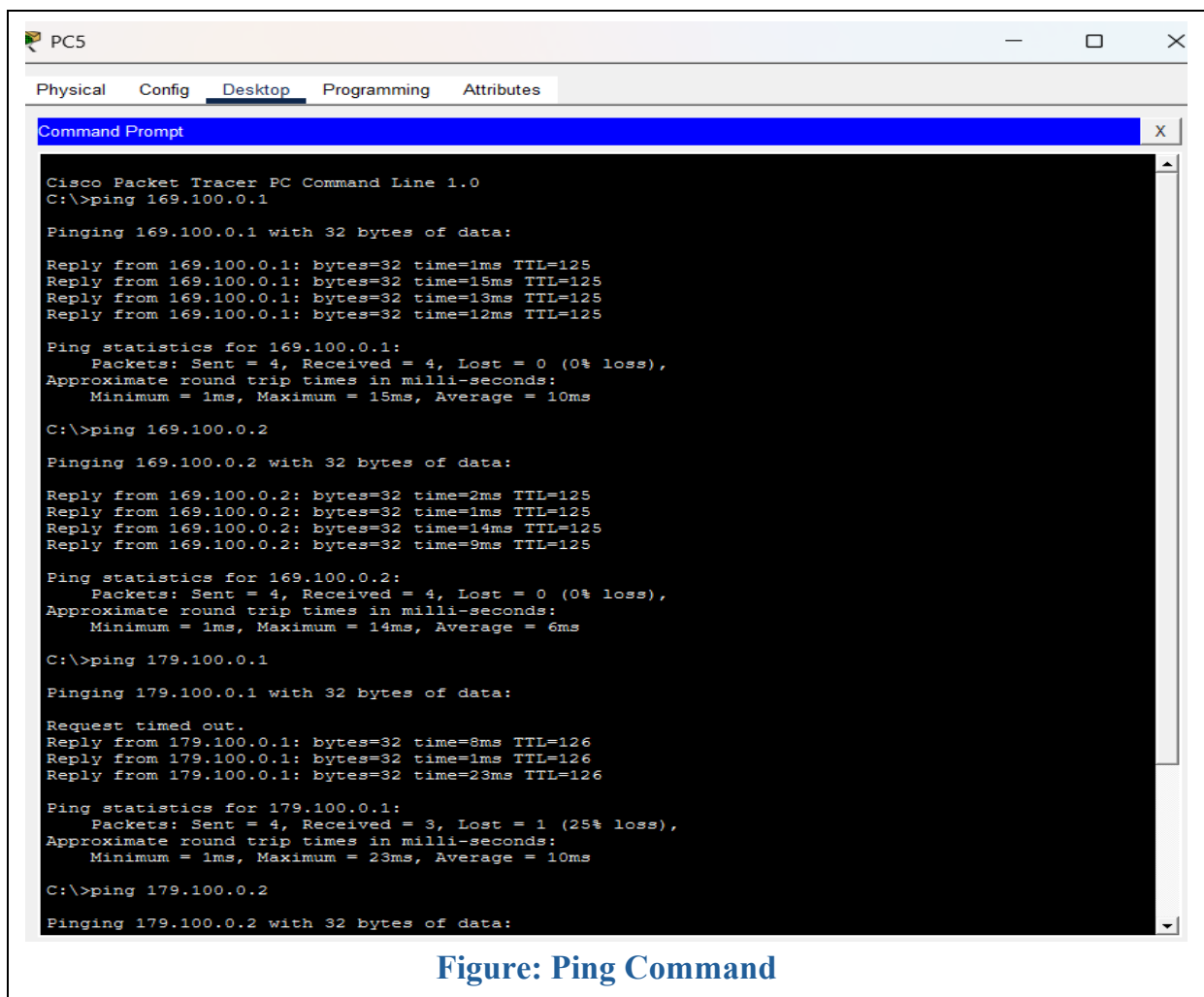


Figure: Ping Command

Test Results Between PCs

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC2	ICMP	Green	0.000	N	0	(edit)	(delete)
	Successful	PC3	PC4	ICMP	Orange	0.000	N	1	(edit)	(delete)
	Successful	PC5	PC1	ICMP	Purple	0.000	N	2	(edit)	(delete)
	Successful	PC4	PC2	ICMP	Blue	0.000	N	3	(edit)	(delete)

Figure: Ping Test Results Between PCs

Conclusion

The network has been successfully designed using Class B IP addressing and configured using static routing. All PCs across different networks can fully communicate with each other, confirming the correctness of routing and addressing.